

NATIONAL EXAM OF HIGHER NATIONAL DIPLOMA – 2021 SESSION

Specialty/Option: *Software Engineering*

Paper: *Practice of Computer*

Duration: *5 hours*

Credit Value: *08*

/50 marks

SECTION A: ALGORITHM AND PROGRAMMING

In number theory, a perfect number is a positive integer that is equal to the sum of its proper positive divisors, that is, the sum of its positive divisors excluding the number itself (also known as its aliquot sum). Equivalently, a perfect number is a number that is half the sum of all of its positive divisors (including itself).

For example, the first perfect number is 6. Its proper divisors are 1, 2, and 3, and $1 + 2 + 3 = 6$. Equivalently, the number 6 is equal to half the sum of all its positive divisors: $(1 + 2 + 3 + 6) \div 2 = 6$.

Using the appropriate software in the computer assigned to you; Answer the following questions:

- Q1. Write a C function `isPerfect(int n)` which check if a number is perfect or not. (10 marks)
- Q2. Write a C program called `Perfect`, prompting the user to Enter a number a positive number and use the function `isPerfect(int n)` to check and display whether the number is perfect or not. (20 marks)
- Q2. Modify the program in such a way that the user enters a positive upper limit "N" and the program print all Perfect numbers between 1 to N (20 marks)

Note: Your program should repeatedly prompt the user to enter a positive number if it is not the case

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (MYSQL) /20 marks

A business company is having the following tables on their database:

Sample Table – Worker

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
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001	Monika	Arora	100000	2014-02-20 09:00:00	HR
002	Niharika	Verma	80000	2014-06-11 09:00:00	Admin
003	Vishal	Singhal	100000	2014-02-20 09:00:00	HR

Sample Table – Bonus

WORKER_REF_ID	BONUS_DATE	BONUS_AMOUNT
1	2016-02-20 00:00:00	5000
2	2016-06-11 00:00:00	3000
3	2016-02-20 00:00:00	4000

Task 1: Using the appropriate software, create the database of the company entitled “COMP”

Task 2: Create the tables “Worker” and “Bonus” including all their columns

Task 3: Insert all Data’s found on “Worker” and “Bonus” tables

Task 4: Create a SQL query to fetch “FIRST_NAME”, “DEPARTMENT”, “BONUS_AMOUNT” from “Worker” and “Bonus” table

SECTION C: WEB DESIGN

/15 marks

A prime number (or a prime) is a natural number greater than 1 that cannot be formed by multiplying two smaller natural numbers. A natural number greater than 1 that is not prime is called a composite number. For example, 5 is prime because the only ways of writing it as a product, 1×5 or 5×1 , involve 5 itself. However, 6 is composite because it is the product of two numbers (2×3) that are both smaller than 6.

1. What do we need to write and run PHP code? (2 marks)
2. Using the appropriate software, write a PHP or JavaScript function taking as parameter a number and check if the number is a prime number or not (5 marks)
3. Given the figure bellow:

LIST OF PRIME NUMBERS

Enter a Number:

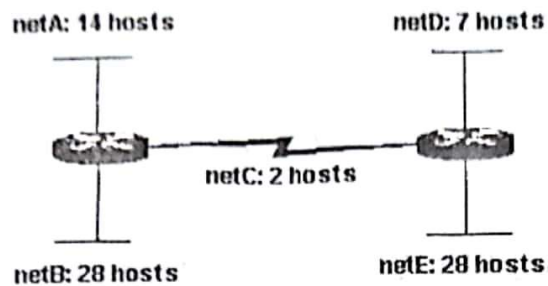
Write now a dynamic web page using the previous function to display the n first prime numbers where n is entered by the user using the HTML form as shown on the figure. (8 marks)

SECTION D: NETWORKING

/15 marks

You have gotten a job as a networking expert to network a newly created company having 5 departments and the manager gives to you the following design:

Figure 3



The manager also said that the IP address given to him by the internet network provider is 204.15.5.0/24.

- 1- List all the tools needed to implement this design (3 marks)
- 2- Which types of cables will be needed? Describe each (2 marks)
- 3- Explain the process of sub-netting and precise the range of each (5 marks)
- 4- Using appropriate software installed in your computer, design the entire network. (5 marks)

Note: Represent just two computers in each sub-network and make sure that you assign the correct IP address to each sub-network